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University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

LAB REPORT

Experiment Number: 05

Experiment Title: Designing a Bluetooth-Based Android-Controlled Robot

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1. Experiment Name

Designing a Bluetooth-Based Android-Controlled Robot.

2. Objective

The main objective of this experiment is to design and implement a Bluetooth-based robot that can be controlled wirelessly using an Android smartphone. The robot uses an Arduino UNO as the main controller, an HC-05 Bluetooth module for wireless communication, and an L293D motor driver to control the movement of two DC motors.

The specific objectives are:

- To establish wireless communication between an Android smartphone and an Arduino UNO.
- To control the forward, backward, left, and right movements of the robot.
- To stop the robot using commands sent from the Android application.
- To understand the applications of microcontrollers, Bluetooth communication, and motor drivers in robotics

3. Apparatus / Hardware & Software Requirements

- List all required tools and components:

Hardware Requirements:

- Arduino UNO – 1
- HC-05 Bluetooth module – 1
- L293D motor driver IC/module – 1
- DC geared motors – 2
- Robot chassis with wheels – 1
- Android smartphone with a Bluetooth controller application – 1
- Battery or suitable motor power supply – 1
- Jumper wires
- Breadboard (if required)
- USB cable for programming the Arduino

Software Requirements

- Arduino IDE
- Tinkercad Circuits (for simulation/design)
- Arduino programming language (C/C++)

4. Circuit Diagram / Schematic

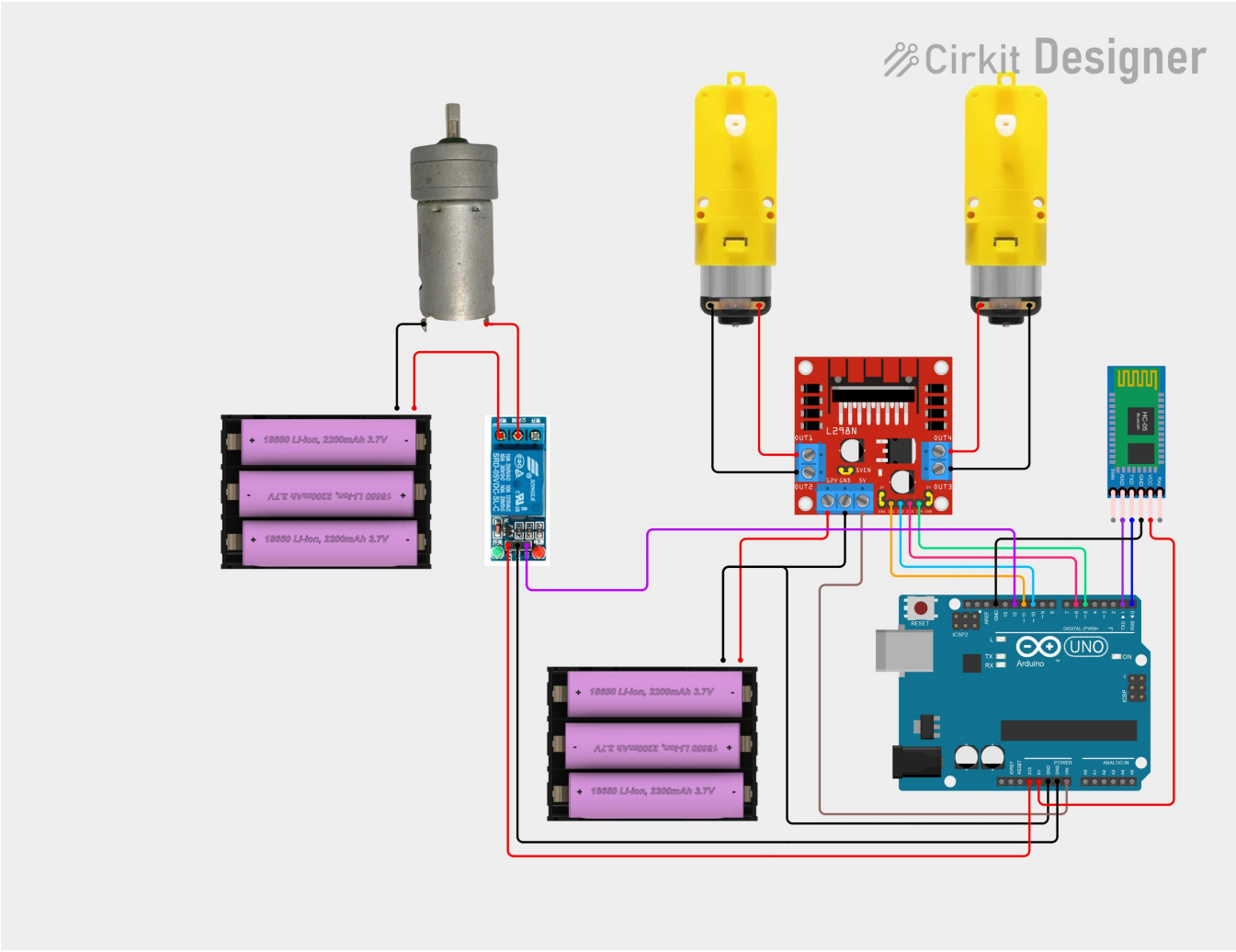


Fig: Bluetooth-Based Android-Controlled Robot

5. Code / Assembly Program

```
#include <SoftwareSerial.h>

// Bluetooth communication
// Arduino D2 = RX, Arduino D3 = TX
SoftwareSerial bluetooth(2, 3);

// Motor driver pins
const int ENA = 5;
const int IN1 = 8;
const int IN2 = 9;

const int ENB = 6;
const int IN3 = 10;
const int IN4 = 11;

char command;
int motorSpeed = 180;

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(ENB, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  Serial.begin(9600);
  bluetooth.begin(9600);

  stopRobot();

  Serial.println("Bluetooth Robot Ready");
}

void loop() {
  if (bluetooth.available()) {
    command = bluetooth.read();

    Serial.print("Received Command: ");
    Serial.println(command);

    switch (command) {
      case 'F':
      case 'f':
        forward();
        break;

      case 'B':
      case 'b':
        backward();
        break;

      case 'L':
```

```
      case 'l':
        turnLeft();
        break;

      case 'R':
      case 'r':
        turnRight();
        break;

      case 'S':
      case 's':
        stopRobot();
        break;

      default:
        stopRobot();
        break;
    }
  }
}

void forward() {
  analogWrite(ENA, motorSpeed);
  analogWrite(ENB, motorSpeed);

  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void backward() {
  analogWrite(ENA, motorSpeed);
  analogWrite(ENB, motorSpeed);

  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);

  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void turnLeft() {
  analogWrite(ENA, motorSpeed);
  analogWrite(ENB, motorSpeed);

  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);

  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void turnRight() {
  analogWrite(ENA, motorSpeed);
  analogWrite(ENB, motorSpeed);
```

```
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void stopRobot() {
  analogWrite(ENA, 0);
  analogWrite(ENB, 0);

  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}
```

6. Output / Observations

After uploading the program to the Arduino, the robot behaves according to the distance measured by the HC-SR04 ultrasonic sensor.

Observed Behavior:

Bluetooth Command	Robot Action
F	Move forward
B	Moves backward
L	Turn left
R	Turns right
S	Stop

The Android application must send these single-character commands. If the robot turns in the opposite direction, interchange the corresponding motor terminals or adjust the motor control logic.

Working Procedure

1. The Arduino UNO, HC-05 Bluetooth module, L293D motor driver, and DC motors were connected according to the circuit diagram.

2. The Arduino code was uploaded using the Arduino IDE.

3. The HC-05 module was paired with an Android smartphone.

4. Movement commands were sent through a Bluetooth controller application.

5. The Arduino processed the commands and controlled the motors to move forward, backward, left, right, or stop.

6. Finally, all commands were tested to verify the robot's operation.

7. Result

The Bluetooth-based Android-controlled robot was designed using an Arduino UNO, an HC-05 Bluetooth module, an L293D motor driver, and two DC motors. The system was programmed to receive wireless commands from an Android smartphone and perform the corresponding movements.

The experiment demonstrates how Bluetooth communication can be used to control a robotic vehicle wirelessly. The objective is achieved when the robot responds correctly to the forward, backward, left, right, and stop commands.

8. Conclusion

In this experiment, a Bluetooth-based robot was designed and programmed using an Arduino UNO. The HC-05 module provided wireless communication between the Android smartphone and the microcontroller, while the L293D motor driver controlled the DC motors.

Through this experiment, the working principles of Bluetooth communication, motor control, PWM-based speed adjustment, and microcontroller programming were understood. This project provides a basic foundation for developing more advanced wireless robotic systems and mobile-controlled automation applications.